

Safety Data Sheet

Effective Date 01/04/2017

1. Identification

Material Name : Ethyl 2-methylacrylate, Ethyl methacrylate
Synonyms : EMA
CAS Number : 97-63-2
Product Use : Coating Resin, Raw Material for plastics,
 Fiber Treatment Agent, Adhesive etc.
Manufacturer / Supplier : **Mitsubishi Chemical Corporation**
 Address : 1-1, Marunouchi 1-chome, Chiyoda-ku, Tokyo, Japan
 Section : MMA Planning Department
 Telephone number : +81-3-6748-7540
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 (0900 – 1700 Japan office hours only).

2. Hazards Identification
GHS Classification:

Health	Environmental	Physical
Acute Toxicity – Not Classified (oral)	Hazardous to the aquatic environment (Chronic) – Not Classified	Flammable Liquid
Acute Toxicity – Not Classified (dermal)		Category 2
Acute Toxicity – Category 4 (inhalation)		Pyrophoric liquid
Skin Corrosion/ Irritation - Category 2		– Not Classified
Eye Corrosion/ Irritation - Category 2B		
Skin Sensitization - Category 1		
Specific target organ systemic toxicity (Single) – Category 3		

GHS Label:

Symbols:  
Hazard Statements DANGER! Highly flammable liquid and vapour

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Harmful if inhaled

Causes skin irritation

Causes eye irritation

May cause an allergic skin reaction

May cause respiratory irritation

Precautionary Statements

Obtain, read and understand the special instructions.

Wear designated respiratory protection, protective gloves, eye/face protection and protective wears.

Keep away from heat/sparks/open flame/high temperature objects.

No smoking.

Keep container tightly closed.

Take precautionary measures against static discharge.

Use explosion proof electric/ventilation /lightning apparatus.

Use only non-sparking tools.

Do not breathe mist/vapors.

Use only in a well-ventilated area or out door.

Do not eat, drink or use tobacco when using this product.

Do not take out the soled working wears from working area.

Wash thoroughly after handling.

Counter measures

IF INHALED: If you feel unwell or breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician.

IF ON SKIN: Remove/Take off immediately all contaminated clothing. Wash with plenty of soap and water. If you feel unwell or skin irritation or rash occurs, get medical advice/attention.

IF IN EYES: Rinse cautiously with water for over 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician. Do NOT induce vomiting.

IF feeling unwell or experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.

IF exposed or concerned: Get medical advice/attention.

Collect spillage.

In case of fire, use dry chemical, carbon dioxide, foam, dry sand for extinction.

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Storage

Storage in tightly closed container at cool and well-ventilated area.

Store locked up.

Dispose

Contents/container is burned in a designated incinerator or are entrusted to qualified waste trader.

Another Hazard out of GHS classification

This material is flammable (flash point is 20.5°C) and vapor may form an explosive mixture with air. (lower flammability limit is 1.8vol%)

High temperature or direct rays may cause polymerization. Polymerization followed exothermic heat and may cause explosive reaction.

3. Composition / Information on Ingredients

Component	CAS Number	Weight %
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Ethyl 2-methylacrylate	97-63-2	99.7%
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EINECS Number

202-597-5

TSCA

Registration

Most important contained substance.

Component	CAS Number	Weight %
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Methyl 2-methylacrylate	80-62-6	0.2%
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EINECS Number

201-297-1

TSCA

Registration

4. First Aid Measures**Eye:** Eye irritation. Flush immediately with large amounts of water for at least 15 minutes.

Eyelids should be held away from the eyeball to ensure thorough rinsing.

If possible, remove contact lenses before flushing the eyes.

Get immediate medical attention.

Skin: Immediately remove contaminated clothing and shoes. Flush the skin with plenty of running water at normal or lukewarm temperatures.

Get immediate medical attention if the appearance of the skin changes or if irritation persists.

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Inhalation: Remove exposed person from source of exposure to fresh air. Wrap in a blanket or similar material to keep warm. Keep quiet. If breathing is weak or stops, loosen clothing, secure an open respiratory airway, and start cardiopulmonary resuscitation (CPR).

Get immediate medical attention.

Ingestion: Get immediate medical attention. Do not induce vomiting unless directed by medical personnel. Doing so will increase the potential for harm. Should be allowed to wash the mouths with water, if the person himself wishes. Do not place anything in the mouth if the person is unconscious.

Most important Sign and Symptoms: In case inhalation, may cause dizziness or choking and a nose or the throat irritation. Contact to eye or skin may cause irritation and a burning by chemicals. If swallowed may cause feeling nausea, vomiting or griping.

Protection of rescuer: In case rescue the victims, wear suitable protective equipments.

5. Fire Fighting Measures

Suitable Extinguishing Media: Use dry chemical, foam, carbon dioxide or dry sand to extinguish fire.

Unsuitable Extinguishing Media: Do not use water in a jet. Doing so may be in danger the spread fire.

Special Hazards: Nothing special item.

Fire-fighting procedures: Keep persons not directly involved in fire-fighting out of the area of the fire. Use dry chemicals, a CO₂ extinguisher, dry sand etc. to extinguish a fire in its early stages. In case of large fires, deprive the fire of a source of oxygen using a foam extinguisher or by other means. Wear protective equipment during fire fighting.

6: Accidental Release Measures

Person protection measures (Protective equipments): Wear protective equipment and operate from upwind during clean up works. Isolate hazard area by rope and deny entry to unnecessary of unprotected personnel.

Environment protection measures: Do not discharge the spill into sewer pipes. If the spill involves small amounts, soak up the material with sawdust, waste cloth or sand, then collect the soaked materials into a container. For extensive spills, block the flow of liquid with earth or sand, cover the liquid surface with foam, then collect as much of the material into an empty container as possible. We recommend using spark proof tools.

Protects from secondary accidents: Promptly remove any flammable materials from

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the surrounding area. Prepare extinguishing media providing ignition.

7. Handling and Storage**Handling:**

- Do not use compressed air to charge, bleed or handle the product.
- If the product is to be handled indoors, provide good ventilation using a local exhaust system or other measures. (In case evaporating at 20 °C atmosphere may be contaminated and the concentration may arrive at poisonous level quickly.)
- Keep the product away from flames, sparks, strong oxidizers and high temperature substances.
- Handle the container gently; do not overturn, drop, strike or drag.
- Wear adequate protective equipment to prevent absorption through the skin, contact with the skin or eyes or inhalation of vapors. -- See Protective equipment.

Storage:

- If the product is to be placed in long-term storage (over three months), inspect the polymerization inhibitor concentrations for significant declines from initial concentrations as necessary. If happen polymerization and heat up, the container should be transferred quickly to well ventilated wide area, and cool down it by large quantity water. After that treat adequate method. In case using bulk tank, store under 30°C (under 20°C is advisable). The space over liquid level in the tank should be controlled to low level oxidation concentration. This treatment may avoid the tank space going in flammable limits. The other side for avoiding polymerization, the oxidation concentration of tank space needs suitable level. So it will be desirable that the tank space is sealed by the gas that oxidation concentration is about 8%.
- Close the container tightly. Store in a cool, dark, well-ventilated place, away from direct sunlight.
- Do not store in the same area with organic peroxides.

Recommended Material: For containers use steel drums or clean drum.

8. Exposure Controls / Personal Protection

Control concentration: Not established.

Exposure Limits: 【Methyl 2-methylacrylate (0.2%)】

ACGIH (2011 version) TLV-TWA : 50ppm

ACGIH (2011 version) TLV-STEL : 100ppm

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Engineering Controls: If the product is used indoors, seal the point of emission or install ventilating equipment or both. Install a safety shower, wash basin and eye bath near the handling area. Indicate the locations of these facilities clearly and prominently.

Personal Protective Equipment (PPE)

Respiratory Protection: Gas mask (to protect against organic gases), airline respirator (in atmospheres with high product vapor concentrations), air respirator, safety goggles or protective shield.

Hand Protection: Avoid skin contact. Wear oil-resistant gloves and boots.

Eye Protection: Wear chemical safety goggles and face shield.

Skin and body Protection: Additional protection may be necessary to prevent skin contact including use of apron, face shield, boots or full body protection.

Occupational Health Measures: Protective Equipment should be inspected periodically. Do not eat, drink or use tobacco when handling this product.

9. Physical and Chemical Properties

Appearance:	Clear, colorless liquid
Odor:	Weak ester odor
Flashpoint:	20.5°C (68.9 °F) (closed cup)
Auto-ignition Temperature:	450°C (842 °F)
Boiling Point:	118°C (244.4 °F) @ 1013hPa
Melting Point:	< -60°C (-76 °F)
Vapor Pressure:	2 kPa @ 20°C
Vapor Density(Air=1):	3.94 (air = 1)
% Solubility in Water:	0.46%
% Solubility of Water:	0.64%
	Soluble in alcohol
Distribution Coefficient(Octanol / Water):	log Pow 1.94
Lower Flammability Limit:	1.8 vol%
Specific Gravity:	0.914/@ 20°C
Viscosity:	0.62 mPa · s/@ 20°C
pH:	N.A.
Molecular Weight:	114.15
Specific heat:	1.93 J/g · °C
Heat of polymerization:	59.1 kJ/mol
Refractive index (n_D²⁰):	1.4157/20°C

10. Stability and Reactivity

Stability: A polymerization inhibitor has been added to the product to stabilize it under normal conditions of handling.

Hazardous Reactivity: Very flammable liquid (flash point is 20.5°C) and the vapor mixed with air generate explosive gas (lower flammability limit is 1.8vol%). Polymerization followed exothermic heat and may cause explosive reaction.

Condition to Avoid: Avoid to storage over the specified restrictions period or temperatures.

Incompatibility: Incompatible with strong acids and peroxide. (Cause of polymerization)

Hazardous Decomposition Products: N.A.

11. Toxicological Information

Acute Oral Toxicity:	LD ₅₀ (Rat)	14,800 mg/kg
Acute dermal Toxicity:	LD ₅₀ (Rabbit)	>5,000 mg/kg
Acute Inhalation Toxicity:	LC ₅₀ (Rat)	8,300 ppm (4H) (vapor)
Skin Corrosion / Irritation:	Short chain methacrylic esters typically show irritating to rabbit skin by long term contact at sealed condition.	
Eye Corrosion / Irritation:	Slight to moderately irritating to eye is observed.	
Respiratory Sensitization:	No information.	
Skin Sensitization:	Positive for skin sensitization in human case. Short chain methacrylic esters are weak skin sensitizer.	
Mutagenicity:	No information.	
Carcinogenicity:	No information.	
Reproductive Toxicity:	No information.	
Target Organ Effects (Single):	Administration test by rat causes disorder of the respiratory system and respiratory tract irritation. Vapor may cause human mucous membrane irritation and central nervous system damage by exposure.	
Target Organ Effects (Repeated):	No information.	

As 0.2% methyl methacrylate,

Respiratory Sensitization: Specified in Respiratory sensitizer group 2 of Japan

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Skin Sensitization:

Society for Occupational Health.

Specified in Skin sensitizer group 2 of Japan Society for Occupational Health.

There is a development example of allergic dermatitis. Specified in EU risk phrase R43 May cause sensitization by skin contact.

Reproductive Toxicity:

Embryotoxicity (early fetal death, reduction of the head butt length and development of hematoma) is reported in dosage of developing maternal toxicity (death, decrease of weight, etc.) by teratogenicity test by rabbit.

12. Ecological Information**Organism Toxicity:**

Daphnia magna acute toxicity NOEC 18mg/L/21day

Persistence / degradability:

No information.

Bioaccumulation:

Bioaccumulation is not expected to be significant.

log Kow = 1.94

Mobility to the soil:

No information.

Hazardousness to the ozone layer:

No information.

13. Disposal Considerations

Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

When incinerating the product, soak up the product using sawdust or similar materials, and burn in small increments. When disposing the products, should entrust to qualified waste trader. We recommend that the Waste Water including these products are treated by activation sludge equipments and are wasted to public space after confirming the quality. When wasting the containers used these products, wash these by water after confirming that the containers do not remain the products.

14. Transport Information**International Regulations :** Follow UN classification**UN Classification :** Class 3 (Flammable Liquid)

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UN Number : 2277**Proper Shipping name :** Ethyl 2-methylacrylate (Stabilized)**Container Classification :** II

Special Safety Measures : Avoid leak the products according to handle the container gently; do not overturn, drop, strike or drag. Watch the temperature of products, especially suddenly temperature raise.

Special Safety Measures and Conditions at Transportation : Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

First Aid Measurements Guide Number : 128P**15. Regulatory Information**

Note: The latest regulatory information should be confirmed with the importer.

16. Other Information**Revision Indicator:** Revised on 01/04/2017

Disclaimer: This SDS was prepared on the basis of materials, data and information available to us at the time of its writing. However, we make no warranty as to the accuracy of the data herein with respect to product content, physical and chemical properties thereof, or associated hazards. The precautions given herein apply to normal handling. For handling under special conditions, users are responsible for implementing adequate safety measures.